

COMMON CONTROL SYSTEM (CCS)

Software User Manual

Software Release 4.2.0 (illustrative) · All platforms edition

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Approval

This manual is approved for release with CCS software release 4.2.0 (illustrative).

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Record of Changes

Each revision adds a row. Change bars in the margin mark text changed since the prior revision.

Rev.	Date	Description of change	Pages affected	Entered by
A	23 Sep 2026	Initial release for CCS 4.2.0 (illustrative)	All	NBS

Safety Summary

These warnings and cautions appear again where they apply. Read them before operating CCS.

WARNING

Do not command a live vehicle from a station whose datalink status is not **LIVE** and green. Commands sent over a degraded link may execute late or not at all. (Sections 5.3.2, 5.6)

WARNING

On lost link, do not re-send route changes until the link is restored and the vehicle has reported its current position. (Section 5.6.1)

CAUTION

Save the mission plan before transferring control. Unsaved changes are lost at handoff. (Section 5.3.2.3)

CAUTION

Do not edit configuration files at the station. Configuration changes are made by the site administrator under CM control. (Section 3.6)

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Preface

Who this manual is for. CCS operators and supervisors at land, ship, and submarine control stations. Site administrators should also use the CCS Site Installation and Configuration Guide.

How to use it. Read Sections 1 through 4 once. Use Section 5 at the station: each procedure is one task, start to finish. Section 5.6 covers emergencies; learn it before you operate a live vehicle.

Platform-specific content. Content that applies to only some vehicles is tagged MQ-25 MQ-4C UUV. Untagged content applies to all platforms. Platform editions of this manual contain only their own tagged content.

Conventions

Table P-1. Typographic and safety conventions

Convention	Meaning
Bold	A control on screen, such as Validate
Monospace	A value you type, or a message the software displays
<i>Italic line after a step</i>	What you should see when the step works
WARNING	Risk of injury, loss of a vehicle, or mission failure
CAUTION	Risk of damage to equipment or loss of data
NOTE	Information that helps you do the task

1 Scope

1.1 Identification

This manual applies to the software identified in Table 1-1.

Table 1-1. Software identification

Item	Value
System	Common Control System (CCS)
Release	4.2.0, build 4.2.0-1187 (illustrative)
CSCIs covered	CCS-CORE, CCS-MPLAN (Mission Planning), CCS-VMGMT (Vehicle Management), CCS-MMGMT (Mission Management)
Edition	All platforms: MQ-25, MQ-4C, UUV

1.2 System overview

CCS is one common software suite that plans, controls, and manages multiple Navy unmanned vehicles. Every vehicle is operated through the same user interface and the same three core functions; only a thin adapter layer differs per platform (Figure 1-1). An operator who learns CCS on one platform can operate another with platform-specific familiarization only.

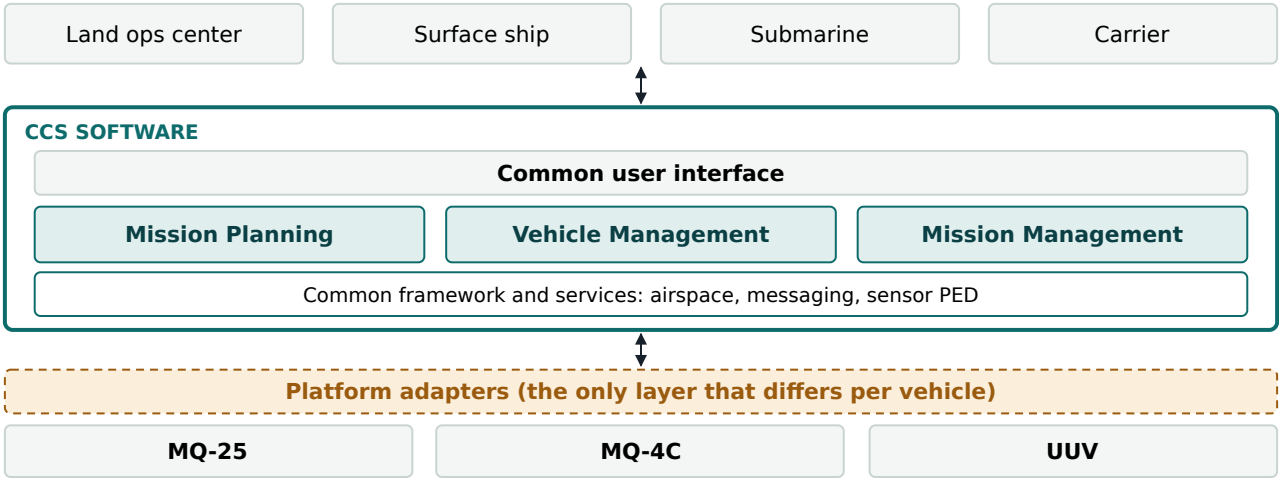


Figure 1-1. CCS system context

1.3 Document overview

Section 2 lists referenced documents. Section 3 summarizes the software. Section 4 covers access. Section 5 is the procedures. Section 6 holds notes; Appendix A holds platform supplements and Appendix B a worked mission.

2 Referenced Documents

Table 2-1. Referenced documents

Document	Purpose
CCS Version Description Document, release 4.2.0	Contents, versions, and known issues for this release
CCS Site Installation and Configuration Guide (CCS-ICG-420)	Installation and site configuration (administrators)
CCS How-To Guides	Task cards for the station, derived from Section 5
CCS Software Requirements Specification	Requirements this release implements
DoD Instruction 5200.48, CUI	Marking of printed outputs and exported data

3 Software Summary

3.1 Software application

CCS lets an operator plan a mission, fly or drive the vehicle, and manage its sensors from one station. Table 3-1 summarizes the three core functions.

Table 3-1. Core functions

Function	What it does	Procedures
Mission Planning	Plan routes, tasks, and timing; validate against airspace, terrain, and reserve constraints; re-plan as things change	5.3.1
Vehicle Management	Establish the link; launch, monitor, command, and recover the vehicle; transfer control between stations	5.3.2
Mission Management	Task sensors; set collection priorities; route sensor products to consumers	5.3.3

3.2 Software inventory

Table 3-2. Software inventory (illustrative)

Component	Contents	Version
CCS-CORE	Framework, user interface, services	4.2.0
CCS-MPLAN	Mission Planning	4.2.0
CCS-VMGMT	Vehicle Management and platform adapters	4.2.0
CCS-MMGMT	Mission Management and sensor PED	4.2.0
Sample plans	Training and smoke-test mission files	4.2.0

3.3 Software environment

CCS runs on a STIG-hardened Linux host with a Java 17 runtime and two displays (illustrative). The site administrator maintains the environment; operators do not change it.

3.4 Software organization and overview of operation

All functions share one main window (Figure 3-1). The numbered areas are described in Table 3-3.

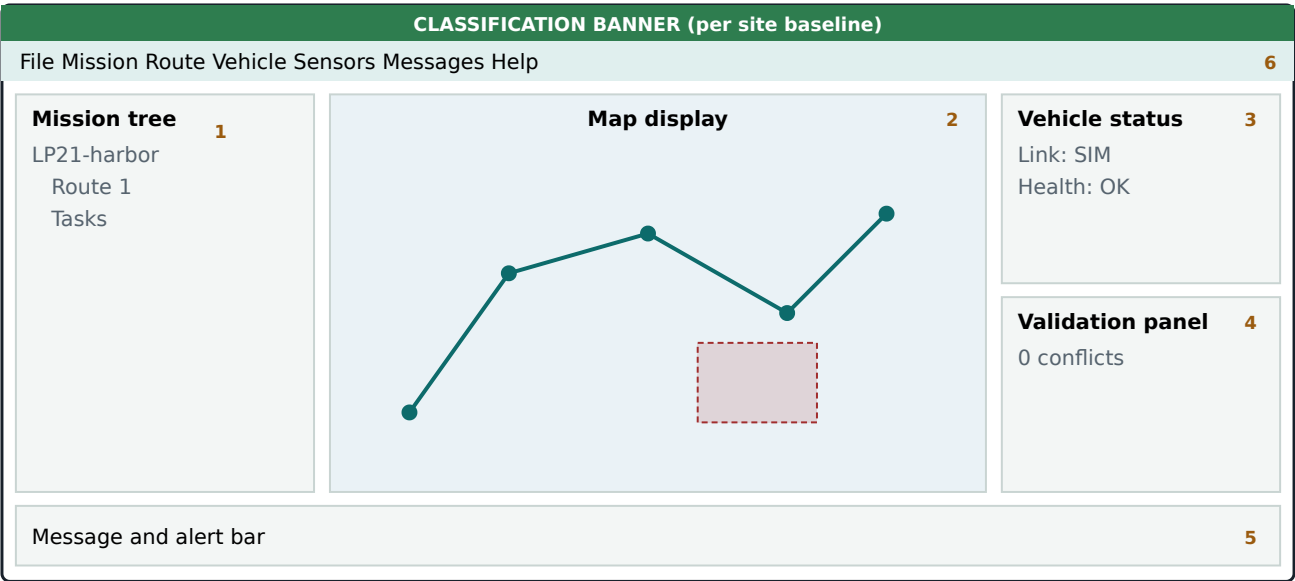


Figure 3-1. CCS main window (illustrative)

Table 3-3. Main window areas

#	Area	What it shows
1	Mission tree	Missions, routes, and tasks in the open plan
2	Map display	Routes, vehicles, airspace, and conflicts; red shading marks restricted zones
3	Vehicle status	Link state and vehicle health at a glance
4	Validation panel	Conflicts found by the last validation
5	Message and alert bar	Alerts, cautions, and system messages (Section 5.7)
6	Classification banner	Set by site configuration; always visible

Table 3-4. Operator roles

Role	Responsibilities
Mission operator	Plans and executes the mission; commands the vehicle
Payload operator	Tasks sensors and manages collection
Supervisor	Approves plans, authorizes control transfer, handles contingencies

3.5 Contingencies and alternate states and modes

Table 3-5. Modes of operation

Mode	Indication	What it means
Normal	Link LIVE , all components healthy	Full operation
Degraded	One non-critical component down	Operate with care; report to supervisor
Lost link	No vehicle telemetry for the platform timeout	Vehicle flies its preplanned lost-link route; see 5.6.1
Simulation	SIM shown in status	Training and verification only; no live vehicle
Failover	Primary station lost	Control moves to backup station; see 5.6.2

3.6 Security and privacy

Log in with your own account; never share credentials. Printed plans and exported files carry the classification banner and must be marked and handled per site procedures. Lock the station (**File** > **Lock**) whenever you leave it.

3.7 Assistance and problem reporting

Report software problems to the site help desk. Include: the time, the station ID, what you were doing, the exact message text from the alert bar, and a screenshot if allowed at your site.

4 Access to the Software

4.1 First-time user

Before first use, complete CCS Operator Basic training and obtain an operator account from the site administrator. Equipment familiarization is covered in training Lesson 1.3.

4.2 Initiating a session

1. At the login screen, enter your user name and password, then select **Log In**.
The main window opens; the classification banner shows at the top.
2. Select **File > Open Mission** and choose the assigned plan.
The mission appears in the mission tree and on the map.
3. Check the vehicle status area: link state and health.
Status matches the mode briefed for this mission (Table 3-5).

4.3 Stopping and suspending work

1. **CAUTION**
Save the mission plan before you log off or hand over. Unsaved changes are lost.

Select **File > Save**.
2. If a vehicle is under your control, transfer control (5.3.2.3) before logging off.
3. Select **File > Log Off**.
The login screen returns.

5 Processing Reference Guide

5.1 Capabilities

Procedures are grouped by function. Each is one task, start to finish; the same tasks appear as CCS How-To Guides at the station.

5.2 Conventions

Table 5-1. Map symbology

Symbol	Meaning
Teal line	Validated route leg
Gray line	Route leg not yet validated
Red line	Leg with a conflict
Red shaded area	Restricted zone
Numbered circle	Waypoint, in route order

5.3 Processing procedures

5.3.1 Mission Planning

5.3.1.1 Create a mission

1. Select **Mission > New**.
2. Enter the mission name and select the platform.
The platform selection sets the limits used by validation.
3. Select **Create**.
The new mission appears in the mission tree.

5.3.1.2 Build and validate a route

A route is the vehicle's travel plan. Validation checks it against every constraint before the vehicle moves (Figure 5-1).

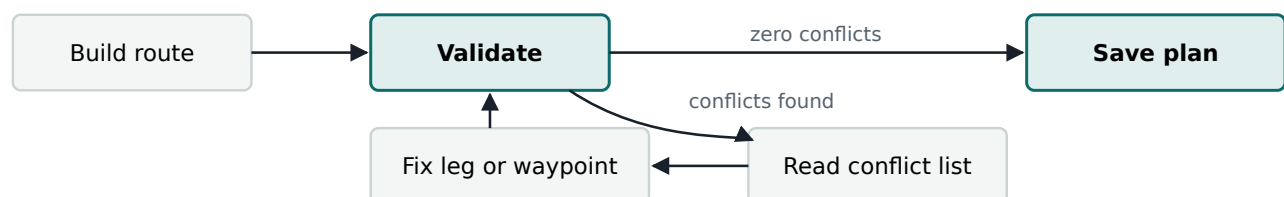


Figure 5-1. Route validation loop

1. Select **Route > New**.
The route editor opens.
2. Click the map to place each waypoint in order.

3. **MQ-25** **MQ-4C** Set altitude and airspeed for each leg.
4. **UUV** Set depth and speed for each leg.
5. Select **Validate**.
The validation panel lists any conflicts; conflicting legs turn red.
6. Correct each conflict (Table 5-2) and validate again until 0 conflicts remain.
7. Select **Save**.

Table 5-2. Common route conflicts

Conflict	Correction
Airspace violation	Move or add a waypoint to route around the zone
Below minimum altitude or depth limit	Adjust the leg to within limits
Speed out of limits	Adjust the leg speed to within platform limits
Insufficient reserve	Shorten the route or remove a waypoint

5.3.2 Vehicle Management

5.3.2.1 Establish the link

1. **WARNING**
Do not command a live vehicle unless link status reads **LIVE** and shows green.

Select **Vehicle > Connect** and choose the assigned vehicle.
*Link status changes to **LIVE**; telemetry updates at least once per second.*

2. Confirm the vehicle position on the map matches the reported position.

5.3.2.2 Monitor vehicle health

Table 5-3. Vehicle status indicators

Indicator	Meaning	Operator action
Green	Normal	None
Amber	Caution condition	Read the alert; notify supervisor
Red	Warning condition	Follow the emergency procedure in 5.6
Gray	No data	Check link; see 5.6.1 if it persists

5.3.2.3 Transfer control

1. **CAUTION**
Save the mission plan before transferring control.

Select **Vehicle > Transfer Control** and choose the receiving station.

2. Wait for the receiving operator to accept.
*Status shows **CONTROL: STATION-02**; your controls turn read-only.*

3. **UUV** Confirm the vehicle has reached the handoff depth before accepting.

5.3.3 Mission Management

5.3.3.1 Task a sensor

1. Select the task in the mission tree, then **Sensors > Assign**.
2. Choose the sensor and collection mode, then set start and end waypoints.
3. Select **Apply**.

The tasked leg shows a sensor icon on the map.

5.3.4 Airspace, messaging, and sensor PED

Procedures for airspace operations, external messaging, and sensor product dissemination follow the same task pattern and are listed in the How-To Guide catalog.

5.4 Related processing

CCS writes an operations log continuously. Supervisors can export it with **File > Export Log**; exported logs carry the classification banner.

5.5 Data backup

1. Select **File > Export Plan**.
2. Save to the site-approved share; name the file with the mission name and date.

5.6 Recovery from errors, malfunctions, and emergencies

5.6.1 Lost link

WARNING

Do not re-send route changes until the link is restored and the vehicle has reported its current position.

1. Note the time and the vehicle's last reported position.
2. Confirm the vehicle is following its preplanned lost-link route on the map.
3. Notify the supervisor.
4. When status returns to **LIVE**, confirm position before resuming commands.

5.6.2 Station failover

1. If your station stops responding, notify the supervisor immediately.
2. The backup operator selects **Vehicle > Assume Control**.

*Status on the backup station shows **CONTROL: BACKUP**.*

5.7 Messages

Table 5-4. Selected system messages (illustrative)

Code	Message	Operator action
CCS-W-101	Link lost	Go to 5.6.1
CCS-C-210	Plan not saved	Save before transfer or log off
CCS-C-305	Validation required	Validate the route before launch
CCS-I-400	Control transferred	None; your controls are read-only

5.8 Quick-reference guide

BEFORE LAUNCH: link LIVE and green · plan validated, 0 conflicts · plan saved · supervisor approval · lost-link route reviewed

6 Notes

6.1 Acronyms and abbreviations

Table 6-1. Acronyms

Acronym	Meaning
CCS	Common Control System
CSCI	Computer Software Configuration Item
CUI	Controlled Unclassified Information
PED	Processing, Exploitation, and Dissemination
SUM	Software User Manual
UUV	Unmanned Undersea Vehicle
VDD	Version Description Document

Appendix A — Platform Supplements

Differences an operator must know when moving between platforms. Limits for each platform are in the platform edition of this manual.

Table A-1. Platform differences (illustrative)

Item	MQ-25	MQ-4C	UUV
Vertical reference	Altitude	Altitude	Depth
Typical control station	Carrier	Land ops center	Ship or submarine
Launch and recovery	Shipboard	Land base	Surface or submerged
Route legs set	Altitude, airspeed	Altitude, airspeed	Depth, speed

Appendix B — Worked Mission Example

A harbor-patrol mission from plan to recovery, using the procedures in Section 5.

1. Create mission LP21-harbor (5.3.1.1).
2. Build a five-waypoint route; validation finds one airspace conflict; add a waypoint to route around it; validate to 0 conflicts (5.3.1.2).
3. Assign the sensor to legs 2 through 4 (5.3.3.1).
4. Establish the link and confirm position (5.3.2.1).
5. Launch, monitor health (5.3.2.2), and recover.
6. Export the plan and log (5.4, 5.5).

Production Notes

How this manual is built, for reviewers and future maintainers.

Aspect	Implementation
Source format	DITA 1.3 topics: concept (Sections 1, 3), task (Sections 4, 5), reference (tables, messages), troubleshooting (5.6)
Structure	<code>ccs-sum.bookmap</code> : one submap per SUM section, following DI-IPSC-81443A
Reuse	Warnings and cautions written once and conref'd; the Safety Summary is built from the same source
Filtering	<code>@platform</code> on tagged steps; this all-platforms edition shows the tags, platform editions drop other platforms
Variables	Release, build, and document number held as keys; updated once per release
Companion outputs	How-To Guides and training job sheets reuse the Section 5 task topics

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